

AMA2111 Mathematics I
1 Complex Number

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- §1 Algebra (DSE)
- §2 Geometric Representation
- §3 Polar Form
- §4 n th root

Some notations:

i : **imaginary unit**, a symbol satisfying $i^2 = -1$

$$\begin{cases} z = x + iy: \text{complex number, } x, y: \text{two real numbers} \\ x = \operatorname{Re}(z): \text{real part of } z \\ y = \operatorname{Im}(z): \text{imaginary part of } z \end{cases}$$

$$\Rightarrow z = \operatorname{Re}(z) + i\operatorname{Im}(z)$$

$\mathbb{C}$: the set of all complex numbers

$\mathbb{R}$: the set of all real numbers

$$z_1 = z_2 \Leftrightarrow \begin{cases} \operatorname{Re}(z_1) = \operatorname{Re}(z_2), \\ \operatorname{Im}(z_1) = \operatorname{Im}(z_2). \end{cases}$$

Definition

Let $a, b, c, d \in \mathbb{R}$ be four real numbers. For two complex numbers $z = a + ib$, $w = c + id$, we define their **sum** as

$$z + w = (a + c) + i(b + d),$$

their **difference** as

$$z - w = (a - c) + i(b - d),$$

and their **product** as

$$zw = (ac - bd) + i(ad + bc).$$

For $z \neq 0$, we define the power as

$$z^0 = 1 \text{ and } z^n = zz^{n-1}$$

for any positive integer n .

Definition

If $a + ib \neq 0$, then the **reciprocal** of z is defined as

$$\frac{1}{z} = \left(\frac{a}{a^2 + b^2} \right) + i \left(\frac{-b}{a^2 + b^2} \right).$$

We define the **quotient** of two complex numbers w and z as

$$\frac{w}{z} = w \frac{1}{z} = \left(\frac{ac + bd}{a^2 + b^2} \right) + i \left(\frac{ad - bc}{a^2 + b^2} \right).$$

If m is a negative integer, we define $z^m = \frac{1}{z^{-m}}$.

From the above definitions, we have

$$z \frac{1}{z} = 1, \quad z^{p+q} = z^p z^q$$

for any integers p and q .

And we have

$$1 + z + z^2 + \cdots + z^n = \frac{1 - z^{n+1}}{1 - z}.$$

The sum $1 + z + z^2 + \cdots + z^n$ is called a **geometric progression with common ratio z** .

Example 1

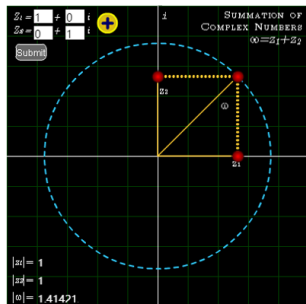
Let $z_1 = 5 - 2i$ and $z_2 = 1 + \frac{\sqrt{3}}{4}i$. Calculate $z_1 + z_2$ and $z_1 z_2$.

Example 2

Let $z_1 = \sqrt{3} + i$ and $z_2 = \sqrt{3} - 2i$. Determine $\frac{z_1}{z_2}$.

§2 Geometric Representation

We may identify a complex number $z = x + iy$ with the point (x, y) in the Cartesian plane $\mathbb{R}_2$. As such, every complex number may be represented by a point in $\mathbb{R}_2$, commonly known as the complex plane or the Argand plane. The complex number 0 is represented by the origin of the complex plane. Since each real number may be considered a complex number with zero imaginary part, the x -axis of the complex plane is known as the real axis, while the y -axis is called the imaginary axis. The real number 1 is identified with the point $(1, 0)$, while the imaginary number i is identified with the point $(0, 1)$.



Conjugate

Definition

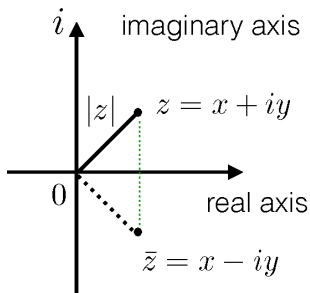
If $z = x + iy$ ($x, y \in \mathbb{R}$), then the **conjugate** of z is defined as $x - iy$. We use the symbol $\bar{z}$ to denote the conjugate of z .

From the definition, we easily obtain

$$\operatorname{Re}(z) = \frac{z + \bar{z}}{2} \text{ and } \operatorname{Im}(z) = \frac{z - \bar{z}}{2i}.$$

Moreover,

$$z\bar{z} = (x + iy)(x - iy) = x^2 + y^2.$$



Proposition

Let z and w be two complex numbers.

- ① $\overline{\bar{z}} = z$
- ② $\overline{z \pm w} = \bar{z} \pm \bar{w}$
- ③ $\overline{zw} = \bar{z}\bar{w}$
- ④ $\overline{\left(\frac{z}{w}\right)} = \frac{\bar{z}}{\bar{w}}$ if $w \neq 0$

The proofs can be done by direct computation. For instance if $z = a_1 + ib_1$ and $w = a_2 + ib_2$, then

$$\overline{z + w} = (a_1 + a_2) - i(b_1 + b_2) = (a_1 - ib_1) + (a_2 - ib_2) = \bar{z} + \bar{w}.$$

The rest of the proofs are left as exercises.

Definition

We define the **modulus** of a complex number $z = x + iy$ ($x, y \in \mathbb{R}$) by

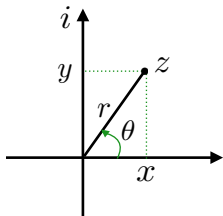
$$|z| = \sqrt{z\bar{z}} = \sqrt{x^2 + y^2}.$$

Proposition

Let z and w be two complex numbers.

- ① $|z| \geq 0$ and $|z| = 0 \Leftrightarrow z = 0$
- ② $|\bar{z}| = |z|$
- ③ $z\bar{z} = |z|^2$
- ④ $\frac{1}{z} = \frac{\bar{z}}{|z|^2}$
- ⑤ $|zw| = |z||w|$
- ⑥ $\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$

§3 Polar Form



$$r = \sqrt{x^2 + y^2}$$

$$\theta : \cos \theta = \frac{x}{r}, \quad \sin \theta = \frac{y}{r}$$

Polar coordinates: (r, θ)

Polar form for a nonzero complex number $z = x + iy$:

$$z = r(\cos \theta + i \sin \theta).$$

r is the modulus $|z|$ of the complex number z . θ is called the **argument** of z . We denote $\theta = \arg(z)$. $\theta = \arg(z)$ is not uniquely defined. In fact, $\arg(z)$ is determined only up to an integral multiple of 2π .

Example 3

Write $\sqrt{3} + i$, and -3 in polar form.

The Complex Exponential Function

Definition

Given $w = x + iy$ for $x, y \in \mathbb{R}$, we define

$$e^w = e^{x+iy} = e^x e^{iy} = e^x (\cos y + i \sin y).$$

This is known as the **complex exponential function** of a complex variable w .

Properties

- 1 If w is real, i.e., $w = x + i0$, then $e^w = e^x$.
- 2 If w is purely imaginary, i.e., $w = 0 + iy$, then

$$e^w = e^{iy} = \cos y + i \sin y.$$

Using this, the polar form of a complex number z may be written as

$$z = r[\cos \theta + i \sin \theta] = re^{i\theta}.$$

The Complex Exponential Function

Recall the following Maclaurin series representation for the functions

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

Then a reasonable definition for allowing imaginary number inputs yi would be:

$$\begin{aligned} e^{yi} &= \sum_{n=0}^{\infty} \frac{(yi)^n}{n!} \\ &= 1 + yi + \frac{(yi)^2}{2!} + \frac{(yi)^3}{3!} + \frac{(yi)^4}{4!} + \dots \\ &= 1 + yi - \frac{y^2}{2!} - \frac{y^3}{3!}i + \frac{y^4}{4!} + \frac{y^5}{5!}i + \dots \\ &= \left(1 - \frac{y^2}{2!} + \frac{y^4}{4!} - \frac{y^6}{6!} + \dots\right) + \left(y - \frac{y^3}{3!} + \frac{y^5}{5!} - \frac{y^7}{7!} + \dots\right)i \\ &= \cos y + i \sin y \end{aligned}$$

Theorem

If $z_1 = r_1 e^{i\theta_1} = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2 e^{i\theta_2} = r_2(\cos \theta_2 + i \sin \theta_2) \neq 0$, then

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)} = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

and

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)].$$

Corollary (De Moirve formula)

If $z = r e^{i\theta} = r(\cos \theta + i \sin \theta)$, then

$$z^n = r^n e^{in\theta} = r^n [\cos(n\theta) + i \sin(n\theta)]$$

for any integer n .

Proof: Use sum and difference of angles identities

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B,$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B,$$

and Pythagorean identities: $\sin^2 \theta + \cos^2 \theta = 1$. We have

$$\begin{aligned} z_1 z_2 &= r_1 r_2 \left[\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 \right. \\ &\quad \left. + i(\cos \theta_1 \sin \theta_2 + \sin \theta_1 \cos \theta_2) \right] \\ &= r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]. \end{aligned}$$

The proof of $\frac{z_1}{z_2}$ and the De Moirve formula are left as exercises.

Example 4

Express the product and the quotient of $\sqrt{3} + i$ and $1 - i$ in polar form.

Example 5

Calculate $(1 + i)^{10}$.

§4 n th Root

Definition

Let $n \geq 2$ be an integer and w be a non-zero complex number. A complex number z satisfying $z^n = w$ is said to be an n th root of w .

In fact, n th roots of w are roots of the polynomial equation $z^n - w = 0$. Since a polynomial equation of degree n has at most n roots, it follows that the number of n th roots of a complex number can not exceed n .

Example

It can be easily verified that 1 , $-\frac{1}{2} + i\frac{\sqrt{3}}{2}$ and $-\frac{1}{2} - i\frac{\sqrt{3}}{2}$ are cubic roots of 1 . Since these numbers are distinct, they are all cubic roots of 1 . We use the notation ω to denote $-\frac{1}{2} + i\frac{\sqrt{3}}{2}$. Observe that $\omega^2 = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$.

Theorem

Suppose that the complex number w has modulus r and argument θ . Then n th roots of w are given by

$$z_k = r^{\frac{1}{n}} e^{i\frac{\theta+2k\pi}{n}} = r^{\frac{1}{n}} \left(\cos \frac{\theta+2k\pi}{n} + i \sin \frac{\theta+2k\pi}{n} \right),$$

where $k = 0, 1, \dots, n-1$ and $r^{\frac{1}{n}}$ is the positive n th root of r .

Proof: Let $w = re^{i\theta}$. If $z = Re^{i\phi}$ is an n th root of w , i.e. $z^n = w$, the De Moirve formula implies $R^n e^{in\phi} = re^{i\theta}$. It then follows $R^n = r$ and $n\phi = \theta + 2k\pi$. The periodicity of sine and cosine implies that all distinct roots can be found by taking $0 \leq k \leq n-1$.

Remark

The above theorem tells us that the n th roots of a complex number $w = re^{i\theta}$ are uniformly distributed on a circle centered at the origin with radius $R = r^{\frac{1}{n}}$, and they are vertices of a regular polygon with n sides. To locate the roots on the circle, we rewrite the argument of the roots in the form

$$\frac{\theta + 2k\pi}{n} = \frac{\theta}{n} + k\frac{2\pi}{n}.$$

Then it is evident that the first root has argument $\frac{\theta}{n}$ and the k th root can be obtained by adding $\frac{2\pi}{n}$ to the argument of $(k - 1)$ th root for $1 \leq k \leq n - 1$.

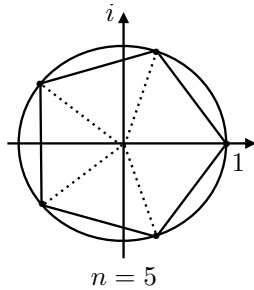
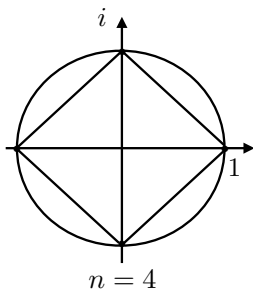
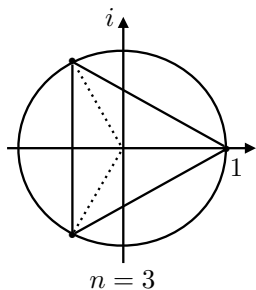
Example 6

Find all cubic roots of $w = 1 + \sqrt{3}i$.

Example

The n th roots of $\omega^n = 1$ are called the n th roots of unity, and are denoted by

$$\omega^k = e^{i\frac{2k\pi}{n}} = \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n}, \quad 0 \leq k \leq n-1.$$



Notice that

$$\omega^0 = 1 = e^{i0}, \quad \omega^{k-1}\omega = e^{i\frac{2(k-1)\pi}{n}}e^{i\frac{2\pi}{n}} = e^{i\frac{2k\pi}{n}} = \omega^k$$

for $k = 0, \dots, n-1$.

We observe that if z_k is one of the n th roots of $w = re^{i\theta}$, then

$$z_k = r^{\frac{1}{n}}e^{i\frac{\theta+2k\pi}{n}} = r^{\frac{1}{n}}e^{i\frac{\theta}{n}}e^{i\frac{2k\pi}{n}} = z_0\omega^k.$$

That is, the n th roots of a complex number can be obtained by a rotation of n th roots of unity followed by a dilation.

Example 7

Find all 5th roots of $w = -32$.